Project Completion Report (PCR)

Project Name	HydraLite: Hydra-Powered Micro-PaaS
Project Number	1400066
Project Start Date	Nov 24, 2025
Project Completion Date	October 16, 2026

1. Context & Original Approach

Cardano Layer 1 consensus settles blocks roughly every 20 seconds. While this ensures high decentralization and security, high-frequency operations—such as micro-payments, batch NFT minting, interactive Web3 gaming, and paywall queries—face throughput bottlenecks and linear transaction fee accumulation if every interaction is committed directly on-chain. The project was undertaken in Fund 14 to remove these barriers by packaging production-grade Layer 2 state channel infrastructure into an accessible, developer-friendly Micro-PaaS engine.

The approach was to build an open-source multi-container architecture combining Cardano Node, Hydra Node, Ogmios, Mithril fast-sync, and an automated scaling daemon, exposed through a standard REST and WebSocket API. The project also included complete GitBook documentation with high-level Mermaid architecture diagrams, professional container security validation, three global community launch workshops, and mainnet deployment templates.

2. Deliverables & Outputs

We successfully delivered the core HydraLite Micro-PaaS platform together with its supporting documentation, testing, security, tutorial, and community-launch outputs. The project is open-source and released under the Apache 2.0 License.

2.1 Core HydraLite Micro-PaaS Engine (Open Source)

- **Multi-Container Orchestration:** A turnkey Docker Compose stack coordinating Cardano Node (10.5.4), Hydra Node (1.2.0), Ogmios (v6), Mithril Client, and the HydraLite FastAPI engine.
- **Source Code:** Publicly hosted on GitHub as `Kwired/Hydra-Powered-Micro-PaaS`.
- **Distribution:** Published verified release image on GitHub Container Registry (`ghcr.io/kwired/hydralite:v1.0.0`).
- **License:** Apache 2.0 License.

2.2 High-Throughput Layer 2 Execution & Auto-Scaling

- Sub-50ms transaction latency inside active Hydra Heads with zero gas fees per transaction.
- Peak throughput benchmarked at **1,060 TPS** with transaction chaining and parallel UTXO workers.
- Fast ledger bootstrapping via Mithril snapshots, reducing initial sync from days to under 30 minutes.
- Automated WebSocket load supervisor that monitors queue pressure and triggers head auto-scaling at 80% load.

2.3 Ecosystem Tooling & Developer Workflow

- Standard REST endpoints (`POST /api/v1/pay`, `GET /api/v1/verify/{id}`) allowing developers to submit state channel transactions without writing custom Haskell or managing UTXOs.
- Persistent WebSockets for live gaming sessions and real-time streaming telemetry.
- Parallel CLI minting tool capable of batch-minting 10,000 CIP-25/CIP-68 NFTs in 9.43 seconds.
- Bring Your Own Keys (BYOK) architecture with read-only key mounts (`:ro`) and zero keys in the public API.

2.4 Documentation, Tutorials & Community Launch

- A public documentation portal was delivered via GitBook and GitHub Pages, featuring interactive Mermaid architecture diagrams, step-by-step setup guides, and complete API specifications.
- Mainnet Deployment Playbook detailing pre-flight security checks, firewall rules, and head-closing contestation procedures.
- Three live community workshops were executed across the AMER, EMEA, and APAC regions with over 160 total live participants.
- The workshops achieved an 88.5% post-workshop survey satisfaction score, and the documentation portal achieved a 94% clarity rating from test users.

2.5 Security Validation

The platform underwent a formal container security and codebase audit conducted by Kush Innovation Labs for release v1.0.0. The audit verified the Bring Your Own Keys (BYOK) model, non-root execution context (UID 1000:1000), supply chain immutability via locked SHA256 image digest, and network segmentation. The final evidence reports a clean result with **zero critical and zero high vulnerabilities** identified in the production release.

3. Usage & Impact

The project targets Cardano dApp developers, Web3 gaming teams, payment gateways, and NFT creators. The platform enables developers to launch high-frequency decentralized applications on Cardano Layer 2 with sub-50ms finality, zero gas fees per transaction inside the channel, and predictable cost structures.

Developer efficiency: Before HydraLite, deploying and operating a Hydra Head required complex Haskell compilation, raw Unix socket bridging, and manual balance reconciliation. The delivered platform reduces Layer 2 integration time from months of specialized engineering to a 10-minute Docker Compose deployment.

Community validation: Over 160 developers attended the global regional workshops, delivering an 88.5% satisfaction score. The GitHub repository received 11 community pull requests for templates, documentation improvements, and CLI enhancements.

Ecosystem benefit: By providing an audited, turn-key Layer 2 Micro-PaaS engine, HydraLite demonstrates how off-chain state channels can solve Cardano's throughput and latency bottlenecks while preserving full Layer 1 security and custody.

4. Sustainability

The project is intended to remain an ongoing open-source platform. The codebase, Docker release images, and documentation are permanently hosted through GitHub and GitHub Pages. Maintenance is managed through community pull requests and GitHub issue tracking.

The stated future roadmap includes browser-based WebAssembly (WASM) client SDKs, dynamic multi-party state channel routing, and integration with future Cardano Hydra protocol upgrades.

5. Evidence & Links

Item	Evidence
GitHub Repository	https://github.com/Kwired/Hydra-Powered-Micro-PaaS
Container Package (GHCR)	https://github.com/orgs/Kwired/packages/container/hydralite/1306676473?tag=v1.0.0
Documentation Portal	https://kwired.github.io/Hydra-Powered-Micro-PaaS/#/
GitBook Navigation & Table of Contents	https://github.com/Kwired/Hydra-Powered-Micro-PaaS/blob/production/v1.0.0/SUMMARY.md
Project Completion Video (Closeout)	https://youtu.be/AKtvavXPbE0
Security Audit Certificate (PDF)	https://github.com/Kwired/Hydra-Powered-Micro-PaaS/blob/production/v1.0.0/AUDIT-CERTIFICATE-v1.0.0.pdf
Mainnet Deployment Playbook	https://kwired.github.io/Hydra-Powered-Micro-PaaS/#/Mainnet_Deployment_Playbook
Community Workshop — Americas (AMER)	https://youtube.com/live/WrZjWepdoNw?feature=share
Community Workshop — Europe/Africa (EMEA)	https://youtube.com/live/8xhsUyZji6E?feature=share
Community Workshop — Asia-Pacific (APAC)	https://youtube.com/live/eKrvjcZDUd4?feature=share

6. Completion Summary

The Fund 14 project delivered a production-ready, security-audited Hydra-powered Micro-PaaS engine, supported by comprehensive GitBook documentation, regional community launch workshops, and active open-source contributions. The completion evidence records 1,060 TPS peak throughput, sub-50ms latency, zero open critical vulnerabilities, 160+ workshop attendees with an 88.5% satisfaction rating, and a 94% documentation clarity score.

This revised PCR follows the same high-level structure and presentation logic as the supplied Project Completion Report for Project 1400098, while using the facts and evidence specific to Project 1400066.